

Prediction of the Quantity and Purity of an Antibody Capture Process in Real Time

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Regulatory recommendations for quality by design instead of quality by testing raise increasing interest in new sensor technologies. An online monitoring system for downstream processes is developed, which is based on an array of online detectors. Besides standard detectors (UV, pH, and conductivity), our chromatographic workstation is equipped with a fluorescence and a mid-infrared spectrometer, a light scattering, and a refractive index detector. The combination of these sensors enables the prediction of specific protein concentration and various purity attributes, such as high molecular weight impurities, DNA and host cell protein content during the elution phase of a chromatographic antibody capture process. Prediction models solely based on online signals are set up providing real-time predictions. No mechanistic models or information about the chromatographic runs is used. These predictions allow online pooling decisions replacing time- and labor-intensive laboratory measurements. Different process variations, such as changes in the column load or elution buffer, are introduced to test the predictive power of the models. Extrapolation of the models worked well when the column load is changed, whereas model adjustment is necessary when the elution conditions are changed considerably.

1. Introduction

In the manufacturing of biopharmaceuticals, both reaching specified quality criteria and lowest possible time to the market are crucial parameters to be focused on. In both cases, process analytical technology (PAT) tools can be beneficial. As recommended by the US Food and Drug Administration (FDA),^[1] the usage of sophisticated sensor systems should be implemented for monitoring and thus better control of the process. Real-time monitoring and model-based prediction of purity and quantity are key steps towards real-time release in the manufacturing of biopharmaceuticals.^[2] According to the ICH guidelines (Q8 R2),^[3] real-time release testing is defined as “the ability to evaluate and ensure the quality of an in-process and/or final drug product based on process data, which typically includes a valid combination of mea-

sured material attributes and process controls.”^[4] In addition, if the product quality can be assessed in real time, hold times for laboratory measurements can be reduced and real-time release approaches become a realistic possibility. Besides that, also in process development and scale-up, online monitoring systems can help to speed up the procedure.^[5–7]

While in upstream processing many different online monitoring systems have been developed in the past few years,^[8–11] fewer approaches have been published for monitoring in the downstream processing (DSP) of biopharmaceuticals. This might be due to measurement time constraints. Actions in DSP have to be placed within seconds, and the measurement has to be suitably fast. At-line high-performance liquid chromatography (HPLC) measurements were proposed by different research groups.^[12–16] However, such HPLC measurements require automated sampling and subsequent data analyses to obtain results fast enough to give feedback information. With respect to automated peak cutting, it would be desirable to obtain results within minutes or even seconds. This limits the application to certain detectors, which fulfill the required specifications. Brestrich et al.^[17,18] as well as Rüdert et al.^[19] demonstrated that multivariate UV/VIS spectroscopy in combination with partial least-squares regression (PLS) modeling is well suited

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for the prediction of different quality parameters in chromatographic applications. In the recent work of Brestrich et al.,^[17] oversaturation of the sensor was prevented by using a flow cell with a variable pathlength. While UV absorption is a well-established method for protein quantification, recording of fluorescence is less frequently used. Excitation of protein's intrinsic fluorescence can be related to a tertiary protein structure. Rathore et al.^[20] published an at-line approach using a fluorescence detector for making pooling decisions during a hydrophobic interaction chromatography step. If coeluting high molecular weight impurities (HMWIs) is the quality aspect of interest, light scattering might be the sensor technique of choice. Li et al.^[21] demonstrated that a multiangle light scattering (MALS) detector can be successfully employed for the characterization of HMWIs during size exclusion chromatography (SEC).

Capito et al.^[22] on the other hand showed that Fourier-transform infrared (FTIR) spectroscopy in the mid-infrared range is also a powerful tool for the prediction of quality parameters in DSP. Capito et al. applied this method at-line using a benchtop device, whereas Großhans et al.^[23] used an attenuated total reflection Fourier-transform infrared (ATR-FTIR) flow cell online for actual real-time applications.

Differently to the discussed approaches, we did not restrict our setup to one sensor, but combined various sensors instead to improve predictability. Precisely, we used an ATR-FTIR, a fluorescence spectrometer, a MALS, and a refractive index (RI) detector in addition to the conventional triple wavelength UV detector and the standard pH and conductivity probe. We focused on sensors, which are commercially available and allow nondestructive measurements within 10 s or less. Moreover, the sensors can be scaled up to pilot or even production scale. In this study, we demonstrate the applicability of the system on the basis of an antibody capture process using protein A chromatography.

To relate the measured online signals to the respective quality attribute of interest, we used PLS regression models. This type of multivariate statistical tool is commonly used for data-based modeling.^[17–19,22,24] It is a linear model relating one (or more) response variables such as quantity or monomer content to several predictor variables such as spectral information from single wavelengths. Typically, the data set is split into training, validation, and test sets in a ratio of 2:1:1.^[25,26] We chose to build the model using eight identical runs. By this design of the data, the model is limited to regular process variations as they occur due to buffer preparation and changes in surrounding conditions such as temperature. The applicability of the models can be broadened by introducing additional process variations such as different loading conditions and changes in the buffer compositions. We have also tested the acceptable range the models can be extrapolated to.

2. Experimental Section

2.1. Cell Culture Supernatant

A monoclonal antibody against tumor necrosis factor alpha was produced in a CHO K1 cell culture. It was cultivated in a

300 mL shake flask preculture and subsequently transferred into a stirred tank reactor with a starting working volume of 10 L using Dynamics AGT medium (Life Technologies, USA) at a temperature of 37 °C. A linear feed rate of 431 g day^{−1} was applied with a glucose concentration of 30 g L^{−1} using Efficient feed A medium (Life Technologies) up to a volume of 30–33% (feed volume/theoretical end volume). The harvested cell culture broth was centrifuged (Avanti JXN-26; Beckmann Coulter, USA) at 1000g for 10 min. The supernatant was aliquoted and stored frozen at −20 °C for further usage. Prior to each chromatographic run, the supernatant was thawed, filtered through a 0.22 µm filter unit (Nalgene, USA) and was kept on ice while loading. On the chromatographic work station, an additional inline filter frit unit was installed to capture possible antifoam or cell debris derived from the fermentation process preventing column clogging.

2.2. Chromatographic Work Station

The chromatographic work station consisted of an ÄKTA Pure 25 System (GE Healthcare, Sweden), which was equipped with two system pumps, sample pump S9 and fraction collector F9-C. It also comprised the standard UV monitor U9-M for triple wavelength detection, as well as the conductivity probe C9 and the pH probe V9-pH. The system was controlled by the software Unicorn Version 6.4. UV absorption at 280 nm, 260 nm, and 214 nm was recorded using a flow cell with an optical pathlength of 2 mm. Light at 280 nm is absorbed by the aromatic amino acid residues of a protein, whereas absorption at 214 nm is related to the peptide bonds and is hence very sensitive. The relation of UV absorption at 280 nm and 260 nm provides an indication of the DNA impurity content.

The work station was upgraded by four additional online sensors: a fluorescence detector, a FTIR spectrometer, a light scattering, and a RI detector as depicted in **Figure 1**. All detectors were connected in series ordered by increasing void volumes of each flow cell and taking into account their pressure sensitivities.

The fluorescence detection setup comprised a laser-induced xenon lamp EQ-99XFC LDLS (Energetiq, USA), a fiber optic multiplexer (Avantes, The Netherlands), a flow cell (FIALab Instruments, USA), and the spectrometer AvaSpec-ULS-TEC with 600 L mm^{−1} grating (Avantes). The multiplexer allowed alternating excitation at 265 nm, 280 nm, 289 nm, 300 nm, 340 nm, and 400 nm. The spectrometer recorded fluorescence counts at a spectral range of 236–795 nm at an integration time of 1 s per excitation wavelength.

The mid-infrared spectrometer MATRIX-FM (Bruker, USA) based on ATR was used to record FTIR spectra at a wavenumber range of 3500–750 cm^{−1}. This detector was cooled with liquid nitrogen to allow high accuracy. FTIR spectra were measured with a resolution of 4 cm^{−1} and 16 scans were averaged per measurement. Prior to each run, a background spectrum of water at the defined flow rate of 1.309 mL min^{−1} was recorded.

The miniDAWN TREOS (Wyatt, USA) was installed to obtain light scattering signals from three angles: 44° (LS1), 90° (LS2), and 136° (LS3). Furthermore, the differential RI detector Optilab T-rEX (Wyatt, USA) was also integrated. Its reference cell was filled with water.

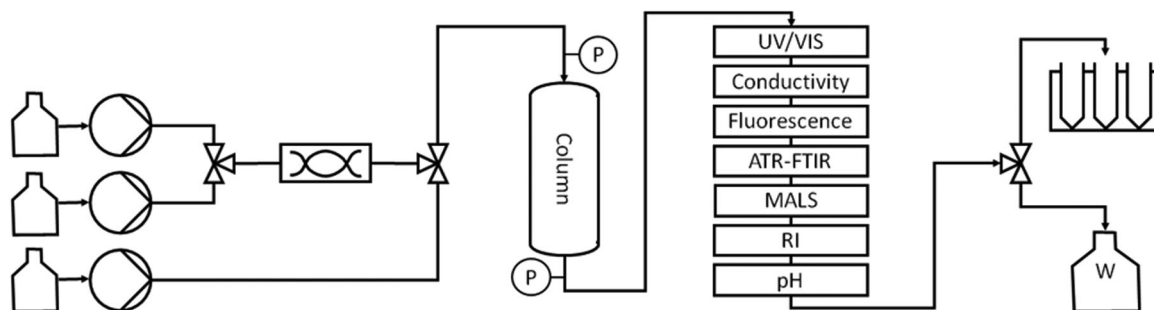


Figure 1. Flow path of chromatographic workstation equipped with additional online sensors (fluorescence, ATR-FTIR, MALS, RI).

All buffers applied were based on aqueous solutions; therefore, water was used as the universal blank solution. As different buffers were used during the method, it would not have been possible to blank all of them especially not mixtures.

The chromatographic run as well as recording of various signals was started through the custom programmed control software XAMiris (Evon, Austria), which provided the central database as well.

2.3. Chromatographic Runs

A Tricorn column (GE Healthcare) with an inner diameter of 10 mm was packed with mAb Select SuRe resin (GE Healthcare) resulting in a column volume (CV) of 12.3 mL. The antibody capture method was conducted at a flow rate of $1.309 \text{ mL min}^{-1}$ and comprised 5 CV equilibration with 20 mM sodium phosphate, 150 mM NaCl, pH 7.4; loading with cell culture supernatant (14.5 mg mL^{-1} resin); 5 CV column wash with 20 mM sodium phosphate, 2 M NaCl, pH 7.4; 7 CV column wash with 20 mM sodium phosphate, 150 mM NaCl, pH 5.0; elution through a gradient over 4 CV changing from pH 5.0 to 100 mM glycine-HCl buffer at pH 3.5, which was then held for another 4 CV to allow complete elution. The column was sanitized with 0.1 M sodium hydroxide, allowing a contact time of 30 min. During elution, fractions of 1.2 mL were collected in a deep well plate. Each well contained 60 μL neutralization buffer (500 mM sodium phosphate, pH 8.0).

For model building, a base data set consisting of eight identical repetitions of the antibody capture process was generated. Additionally, so-called challenging runs were performed to test the predictive strength of the models. Besides a different loading material, also higher column load (24 mg mL^{-1}), as well as variations of elution buffers (glycine buffer at pH 3.0 and 4.0 and sodium acetate at pH 3.6) was tested.

2.4. Concentration Step

The CHO-derived supernatant was concentrated using an Äkta Flux (GE Healthcare) with a Sartocoon Slice 200 Hydrosart membrane, which had a cut-off of 2 kD (Sartorius, Germany). The feed pressure was kept constant at 1.0 bar, and the retentate pressure was adjusted to 0.75 bar, which resulted in a

transmembrane pressure of 0.88 bar. This allowed a flux over the membrane of $\approx 1.5 \text{ mg min}^{-1}$. The supernatant was concentrated twofold.

2.5. Analytics

2.5.1. Antibody Concentration

The antibody concentration was determined by high-performance monolith affinity chromatography using an analytical protein A column as described by Tscheließnig and Jungbauer.^[27] An n-protein A CIM disk (BiaSeparations, Slovenia) was used on an Agilent 1100 HPLC system (Agilent, USA) equipped with a binary pump and a variable wavelength detector. Sodium phosphate (30 mM), 1 M NaCl, pH 7.4, was used as running buffer and 10 mM HCl, pH 2.0, as elution buffer. The method was run at a flow rate of 1 mL min^{-1} . All samples were filtered through a $0.22 \mu\text{m}$ syringe filter prior to measurement. Injection volume was set to 100 μL and absorbance at 280 nm was measured. The method had a standard deviation (STD) of 5%.

2.5.2. Monomer and HMWI Content

Analytical SEC analysis was performed in isocratic mode to separate monomeric antibodies from impurities of higher and lower molecular weight. A TSKgel G3000 SWxl column (Tosoh Bioscience, Japan) was connected to an Agilent 1290 Infinity UPLC system with a quaternary pump and a diode array detector; 150 mM potassium phosphate, pH 6.5, was used as running buffer at a flow rate of 0.4 mL min^{-1} . The injection volume was set to 20 μL and absorbance at 214 nm was measured. Based on the area of the elution peaks, the content of HMWI, monomer, and LMWI was calculated. Determination of monomer had a STD of 5%, whereas HMWI values varied by 10%.

2.5.3. Host Cell Protein (HCP) ELISA

A Nunc MaxiSorp 96-well plate (Thermo Fisher Scientific, USA) was coated with an anti-CHO HCP antibody (produced in goat, 3G-0016-AF; Cygnus, USA) in 0.2 M sodium carbonate buffer pH 9.4 for 2 h at 37°C under slight shaking. Residual binding sites were blocked with 3% bovine serum albumin (BSA) (Sigma Aldrich, USA) in Tris-buffered saline incubating

for 1 h at 37 °C under slight shaking. The CHO HCP standard (F553H; Cygnus) and samples were diluted with sample buffer containing 1% BSA in Tris-buffered saline, transferred into the first row of the plate and serially diluted 1:2 again with sample buffer. On each plate a calibration with the standard was done in duplicates. The plate was again incubated for 1 h at 37 °C under slight shaking before the detection antibody (anti-CHO HCP HRP conjugate, F551C; Cygnus) was added followed by another incubation period of 1 h at 37 °C while shaking. 3,3',5,5'-Tetramethylbenzidine peroxidase EIA substrate (Bio-Rad, USA) was added for staining and incubated for 30 min at room temperature without shaking. The addition of 1 N sulfuric acid stopped the enzymatic reaction. The final absorbance was measured at a wavelength of 450 nm and a reference wavelength of 630 nm using a Tecan infinite 200 Pro plate reader (Tecan, Switzerland). The obtained values in ng mL⁻¹ were related to antibody concentration; thus results are stated in ppm. The method had a STD of 15%.

2.5.4. DNA Concentration

Contamination with double stranded DNA was assessed using the Quant-iT PicoGreen dsDNA Assay Kit (Thermo Fisher Scientific). On each 96-well plate, a calibration with the standard was done in duplicates. Standard and samples were prediluted with 1× TE buffer (10 mM Tris-HCl, 1 mM EDTA, and pH 7.5). A total of 170 µL of the sample was transferred into the first row of the plate, and serially diluted 1:2. 100 µL of each sample was mixed with 100 µL coloring reactant solution and incubated in the dark for 2 min. Fluorescence intensity was measured using a Tecan infinite 200 Pro plate reader (Tecan) at an excitation wavelength of 480 nm and emission at 520 nm (filter with a bandwidth of ±20 nm). The obtained values in ng mL⁻¹ were related to antibody concentration; thus results are stated in ppm. The method had a STD of 20%.

2.6. Modeling

The eight identical repetitions of the antibody capture process were used to set up a statistical prediction model. The online data were stored in the database of the custom programmed control software XAMiris (Evon). In total 15725 input variables were derived from the online sensors per training run delivering data points within seconds. Individual files for the different online devices were extracted with a common time stamp. This was of great value for time alignment of the individual online variables as the time grid of the different online sensors ranged between 1 s and 10 s. The data were read into the statistical computing environment R^[28] where all the statistical analyses were performed. First, quality and consistency checks of the data were performed by visual inspection in 2D as well as 3D for spectral data together with descriptive statistics. Several methods and combinations of signal smoothing, background correction, normalization as well as the use of derivatives with different parameter settings were evaluated. The finally selected preprocessing method combined several methods. For the fluorescence spectra signal smoothing using

the Savitzky–Golay filter^[29] followed by baseline correction and normalization to equal lengths for each time point was chosen. The ATR-FTIR data were background corrected by subtracting an averaged blank spectrum recorded just before elution followed by signal smoothing using the Savitzky–Golay filter and taking the second derivative thereof. All preprocessing methods used are methods that can also be applied in real time as no information about the future of a data point is used.

For model building the time intervals of the online data were reduced by averaging over the 1 min time grid of the offline analytics of the key variables of interest (i.e., concentration and purity).

PLS regression is a well-established modeling technique in chemometrics. PLS transforms the original predictor variables into a set of latent variables, which are also linear functions of the original predictors. These latent variables are then used as predictors in a multiple linear regression model for the response y . Linear combinations are determined such that a maximum covariance between the scores (the values of the latent variables) and the response is achieved.^[30] The number of components (latent variables) is an optimization parameter and was determined within the framework of a leave-one-run-out cross-validation, i.e., one of the eight runs is held out, and predicted by a model fit on the remaining seven runs. This procedure is repeated until each run has served as a test run once. The number of components was limited to avoid overfitting. The output of this algorithm is the optimized model parameter and a prediction error estimate. The root mean square error (RMSE) was calculated according to the following equation:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

where n is the number of observations, y_i represents the measured values, and $\hat{y}_i$ the predicted values.

For PLS scaling is indispensable as all the different variables were measured on different scales and units. In this study, autoscaling was performed, where each variable x was transformed individually by subtraction of the mean of x and division by the empirical STD.

Based on expert knowledge (such as amide bands and fingerprint regions) the spectral data were reduced to certain regions of interest and different combinations of sensor signals were tested. Numerous subsets of predictors (online variables) were selected, prediction models were generated on the eight identical runs using eightfold cross-validation for each of the five responses and RMSEs were computed. The set of predictors yielding the lowest RMSE for a given response was selected for the prediction model of this response. Details about the selected PLS models are given in the Supporting Information. For quantity a variable set consisting of UV, pH, and RI gave the smallest prediction error. The monomer content was predicted best using UV, RI, and fluorescence signals. The final model for HMWI was based on UV, RI, fluorescence, and ATR-FTIR signals. HCP impurities were predicted using UV, RI, MALS, and ATR-FTIR signals. And finally for dsDNA impurities a variable set consisting of UV, RI, MALS, and fluorescence signals was selected.

3. Results

A set of online sensors was used to monitor a chromatographic antibody capture process. PLS models were generated on the basis of eight identical runs relating offline and online data. These models allowed the prediction of antibody concentration, monomer, and HMWI content as well as coeluting DNA and HCP concentration during the course of the elution peak. Predictions in high resolution enabled the visualization of the developing of these quality attributes during the elution step, thus providing the basis for real-time pooling decisions. By applying different process variations such as different column load and elution buffer conditions, the extrapolation ranges of the models were tested.

3.1. Prediction of Test Data Set

The training data set consisted of online signals and corresponding analytical results of the elution fractions obtained from eight repetitions of the antibody capture process. Each offline parameter represented a response variable for the PLS models. Model parameters were optimized on the basis of the training data set. During cross-validation, models with the lowest average RMSE were selected. All models did include the UV signals. This is not surprising, as UV reflects the protein concentration and can basically be related to most parameters. Elution of the antibody was initiated by pH change; thus it is obvious that the product quantity is related to this parameter as well as RI, which is sensitive to buffer changes. Fluorescence signals correspond to protein's tertiary structure. Both monomer and HMWI models comprised fluorescence signals, which might be explained by the structural change of the antibody forming aggregates. Additionally, the HMWI model included also ATR-FTIR data, which also contain structural information of proteins. Prediction of HCP and dsDNA required a complex selection of sensor signals. dsDNA may be present associated with histone and therefore also fluorescence contributed to the prediction.^[31]

As an independent test set we chose variation runs from a different fermentation batch and doubled titer concentration in the supernatant. The direct comparison of the measured and predicted values of the individual fractions from the different runs of the test data set is depicted in **Figure 2**.

Essentially, the antibody concentration during the elution reflected the UV signal showing a Gaussian peak shape (**Figure 2A**). The average prediction error for this parameter over all the six runs was 1.14 g L^{-1} . As seen in the graph, especially in the middle part of the peak, the antibody concentration was slightly underestimated, but generally in good agreement with the reference analytics.

Predicted and measured values of product purity in terms of monomer content as well as content of coeluting impurities such as HWMI, HCPs, and DNA are shown in **Figure 2B–E**. The impurity content—especially DNA—in some fractions at the peak flanks were below the detection limit; thus no values are depicted. In the middle fractions monomer purities of up to 98% were measured, whereas at the peak flanks the impurity content was elevated. HCPs and DNA were also coeluting with

the antibody. However, in relative terms, the HCP and DNA concentration was decreased in the peak center leveling off at $\approx 2200 \text{ ppm}$ and 3 ppm , respectively.

While the monomer content was predicted with an average RMSE of 6.9%, the HMWI prediction resulted in an average RMSE of 2.3%, which is satisfactory. The predictions of HCPs and DNA yielded in average over all fractions RMSEs of 59959 ppm and 19.5 ppm, respectively. These deviations result from the large values at the peak front and rear, where the model prediction was not very accurate. Nevertheless, the trend of the offline data showing an inverse peak shape could be predicted well.

3.2. Pooling Decision

Applying the established models on real-life capture processes and enabling predictions of the quality parameters in a timely manner is the ultimate goal of an online monitoring system. Only if the information is available within seconds, real-time pooling decision can be based on the online sensors.

The computational calculation of our PLS models can be performed within seconds; thus the recorded signals can be processed accordingly fast. In a hypothetical experiment, solely based on the online signals of a new independent run performed on material from a different fermentation batch, the profile of the antibody concentration and the purity and impurities over the course of the elution peak were predicted with high resolution and compared to offline derived values—as illustrated in **Figure 3**. As shown in graph, different modeling approaches (**Figure 3B,C**) were opposed to the offline values (**Figure 3A**). The models gave predictions in a time grid of 1 s compared to the offline measurement time grid of 1 min. In **Figure 3B** the models performing best on the basic set are used, whereas in **Figure 3C** simpler models without spectral data are used. These simpler models showed larger prediction errors compared to the best performing models; however, they also gave less noisy predictions. Details about the selected models are given in the Supporting Information.

We assumed the following pooling criteria: purity in terms of monomer content $>90\%$, HCP concentration $<5000 \text{ ppm}$, DNA concentration $<10 \text{ ppm}$. The pools should meet these criteria on average. Based on this information, the pools were defined slightly differently. Offline, the optimal peak cutting decision can be placed by averaging earlier and later eluting fractions, whereas online the decision has to be taken in real time and future values are still unknown. In our scenarios, the online pool collection started once the predicted value fell below the specified limits and stopped as soon as the average value exceeded the limits. By such a peak integration, the noise in the predictions can be handled easily.

In **Table 1** a comparison of the quality parameters of the different pools is given. Offline pooling is based on offline measured values, whereas the automated pooling decisions are based on model-predicted values, as they are shown in **Figure 3B,C**. The antibody concentration, yield and all purity-related properties are comparable in all pools.

Our simple peak cutting approach would have led to a slightly delayed start of the pool collection in real time, which

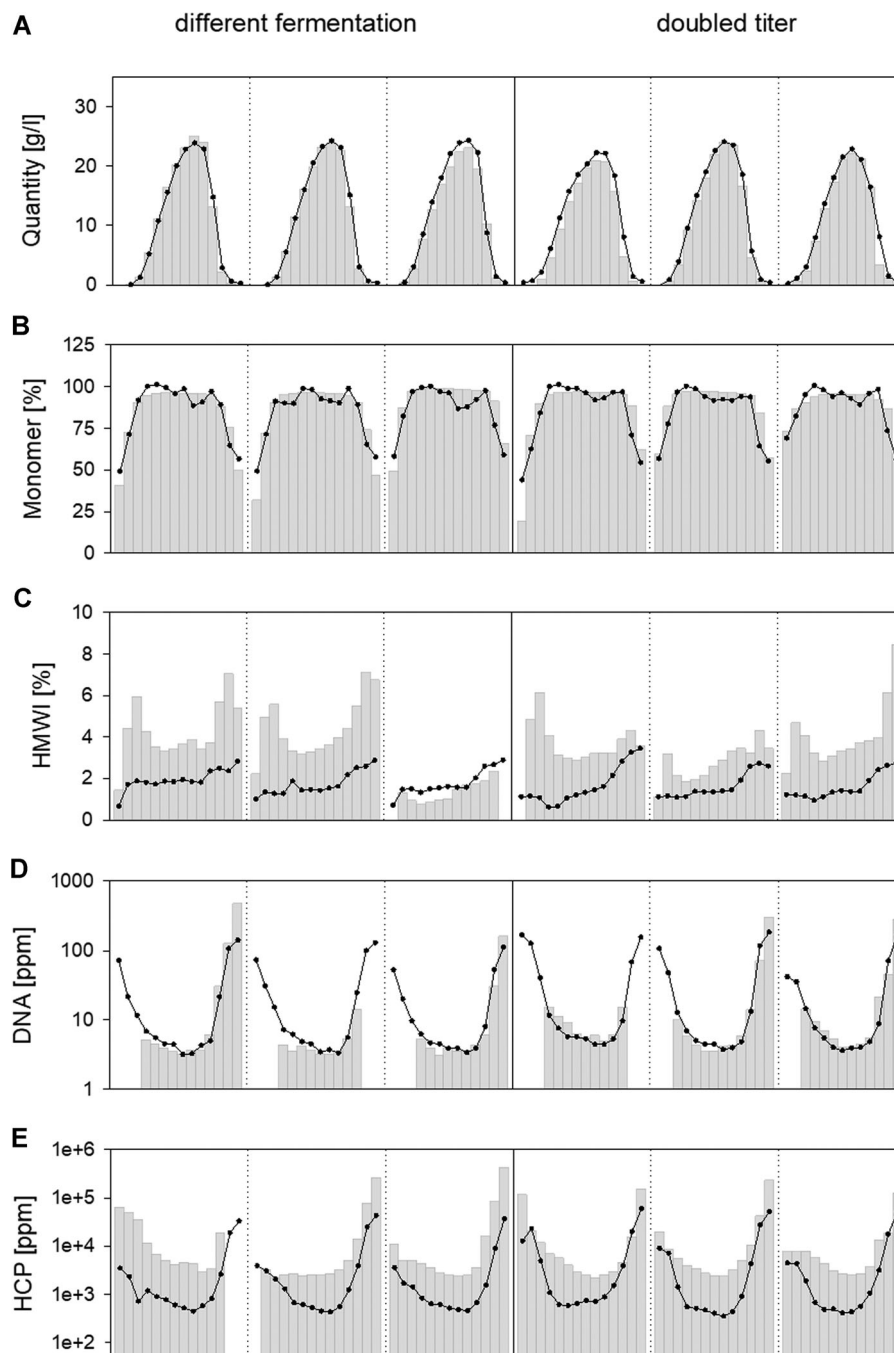


Figure 2. Comparison of the PLS model prediction for (A) antibody concentration (STD $\pm$ 5%), (B) monomer (STD $\pm$ 5%) and (C) HMW content (STD $\pm$ 10%), (D) DNA concentration (STD $\pm$ 20%), (E) HCP concentration (STD $\pm$ 15%) with measured offline values for the test data set. Solid lines represent the predicted values, and bars represent the measured values.

might have further diluted the actual antibody concentration adding up to the lowered predicted antibody concentration. Therefore, more sophisticated tools for the determination of the peak cutting start should be developed. But even with our simple approach we could demonstrate that considering the time and costs, which can be saved by avoiding offline analyses, minor differences in yield can be compensated.

3.3. Model Extrapolation

As the models have been trained on a data set derived from eight repetitions, they are able to handle small process variations due to temperature, minor pH changes, and slight differences in buffer component concentrations. However, the models would be really powerful if they could be applied at process conditions outside of this calibration range as well.

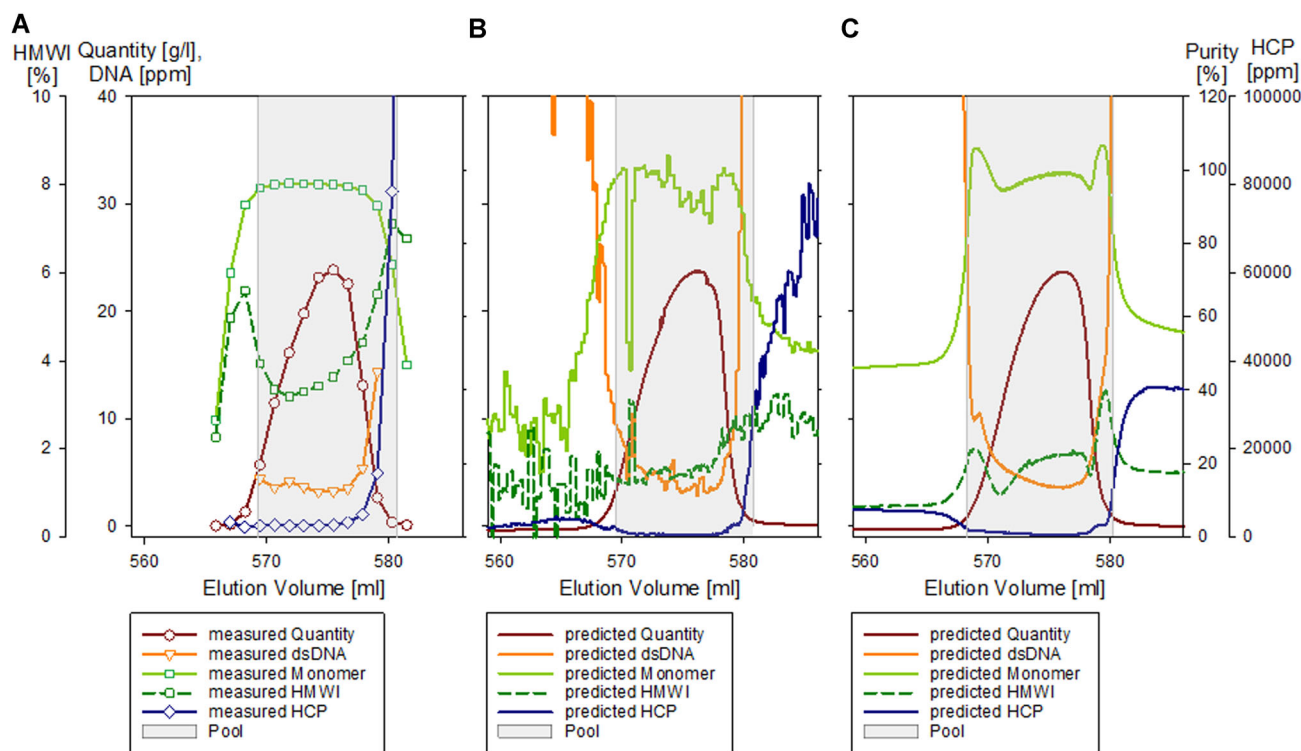


Figure 3. Pooling decision (monomer >90%, HCP <5000 ppm, and DNA <10 ppm) based on (A) offline values and (B,C) online predicted values applying different models.

Additional runs with large variations in process conditions were performed to test the predictive power of the model. More precisely, the following process variations were introduced: increased column load, elution with glycine-HCl at pH 3.0 and pH 4.0, elution with sodium acetate at pH 3.5, and a different antibody.

One parameter was changed at a time and experiments were performed in triplicates. Fractions were collected, analyzed, and compared to the model-derived values, which is illustrated in **Figure 4**. The impurity content—especially DNA—in some fractions at the peak flanks were below the detection limit; thus no values are depicted. The quantity of the antibody could be well predicted by the model, even beyond the ranges the model was trained on. When the elution behavior is changed considerably (pH 3.0 or 4.0 or different buffer component), the monomer model predicted the purity less precisely, but for all other conditions it is also possible to extrapolate this model. With regard to HMWI content, similar conclusions can be

drawn: if the altered elution conditions caused significantly different profiles, the model did not work anymore. This was very pronounced when sodium acetate was used as a elution buffer component. It turned out that the antibody was not stable in this buffer and was prone to aggregation leading to HMWI contents of up to 40%. While HMWI content was raised, less DNA coeluted at pH 4.0 and with sodium acetate, which was overestimated by the model. The supernatant containing a different antibody comprised less impurities as for DNA and HCPs. In this case, the DNA model could not be applied satisfactorily. The HCP model worked moderately well, but the trend indicating a U-shaped elution profile could be clearly seen. Although the actual numbers were underestimated in most cases, pooling might still be possible as demonstrated with our hypothetical experiment. Elution with sodium acetate was again an exception; HCPs could not be predicted at all with this alteration.

4. Discussion

With this study, we demonstrated the proof of principle that an online sensor array combined with PLS modeling can be used for the prediction of multiple quality parameters during a preparative chromatography run. Other studies dealing with online monitoring of an antibody capture process have been published using only one online sensor.^[11,12,14] Großhans et al.^[23] separated an antibody from lysozyme with a cation exchanger and predicted the elution behavior via an FTIR detector yielding an RMSE of prediction of the antibody

Table 1. Pool properties based on offline and automated pooling decision.

	Quantity [g L ⁻¹]	Monomer [%]	HMWI [%]	DNA [ppm]	HCP [ppm]	Yield [%]
Offline pooling	14.0	94.7	4.0	5.0	4086	93.0
Automated pooling 1	14.7	92.7	1.7	9.9	1708	91.9
Automated pooling 2	14.7	97.2	1.8	9.9	1881	91.4

concentration of 2.42 g L^{-1} (max. value 112 g L^{-1}). Rüdert et al.,^[19] on the other hand, predicted antibody breakthrough during capture resulting in an RMSE of prediction of antibody quantity as low as 0.01 g L^{-1} (max. value 3 g L^{-1}) using a UV detector. Brestrich et al.^[18] purified an antibody from HMWI with a cation exchanger and reached a RMSE of prediction as low as 0.014 g L^{-1} for monomer (max. value 1 g L^{-1}), 0.0036 g L^{-1} , and 0.0007 g L^{-1} , respectively for two different HMWI species also based on spectral UV data. We could not achieve such low prediction errors; however, the reported

RMSEs depend on various factors such as the maximum values of concentration, which were significantly lower in the two cases. In our case, antibody concentrations between 0.1 g L^{-1} and 25 g L^{-1} were measured and predicted with an RMSE of 1.14 g L^{-1} which is the average prediction error of 80 fractions in total. Besides, the RMSE is also a matter of the measurement error of the offline analyses, which builds the basis of the models. Our bio-assays exhibit measurement errors of $\pm 5\text{--}20\%$.

Most importantly, the trends in the peak flanks have to be predicted precisely, because this is where peak cutting decisions

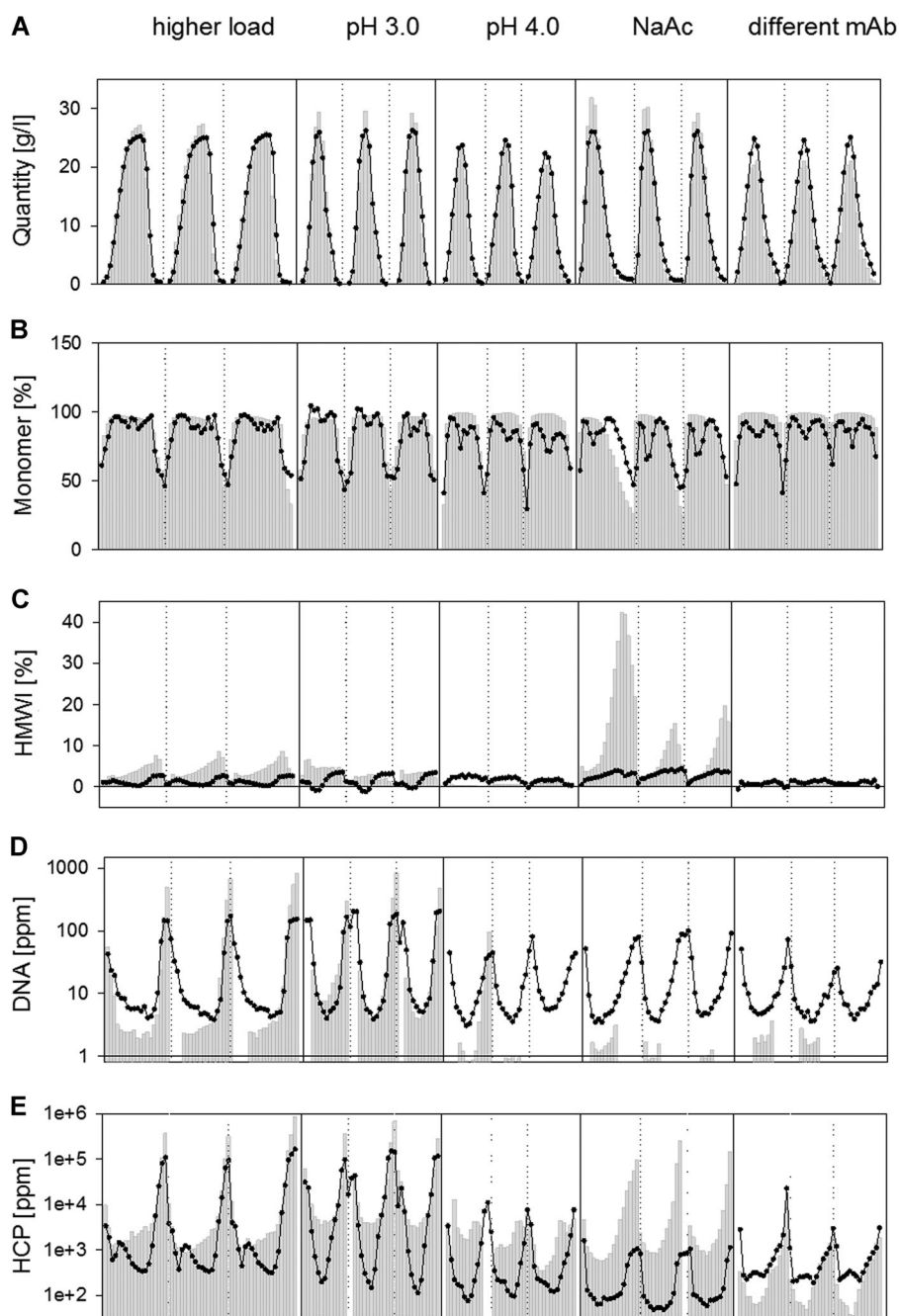


Figure 4. Prediction of quality parameters beyond the model training range. (A) Antibody concentration, (B) monomer content, (C) HMWI content, (D) DNA and (E) HCP. Each condition shows three elution profiles of separate chromatographic runs.

have to be placed. Sensor saturation during the main part of the pool can be compensated by other sensors. The final concentration may deviate slightly from the actual value, but the product meets specified criteria and an optimized yield can be achieved. Other modeling techniques such as random forest, neural networks, or boosted structured additive regression^[26,32] may improve the prediction, but such an assessment was beyond the scope of this work.

Variations in the training data set will reflect process variations, but additional artificially introduced errors will affect the accuracy of prediction. The training data set will not replace experiments to define the design space of a process. Our developed models are able to handle changes in the column load, which may occur due to fluctuations in the upstream process. However, if the chromatographic process is changed severely, e.g., when a completely new elution buffer is used, the training data must be extended to allow reliable predictions. As the models were built based on eight identical repetitions, typically occurring variations in pH and buffer components are within the prediction range. But if the models needed to be applicable for a broader range, additional experiments have to be taken into the training data set.

5. Conclusion

We developed a fast, nondestructive online monitoring system for chromatographic processes, which is commercially available and can be scaled up to manufacturing scale. With this monitoring system comprising seven different online sensors, we demonstrated successfully that several different quality parameters can be predicted in real time simultaneously. Model extrapolations are possible, but are limited to modifications in the column load and small changes in the buffer composition. The predictions can be performed within seconds, thus enabling online pooling decisions. Such PAT tools will enable process control and thus constant product quality without the need of cost- and labor-intensive offline analyses. Another economic aspect is the avoidance of additional hold times between unit operations. From the quality perspective it is also in accordance with FDA's recommendation. It is requested to include on-line sensors to improve the constant quality of the product.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

ATR-FTIR, fluorescence, light scattering, online monitoring, PLS modeling

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